Yufeng Gu

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Education

University of Michigan — **Ann Arbor, MI, USA**
Ph.D., Computer Science and Engineering Department — *Sep. 2020 - Apr. 2026*

Zhejiang University (ZJU) — **Hangzhou, China**
B.Eng., College of Information Science and Electronic Engineering — *Sep. 2016 - June 2020*

Experience

Meta Platform Inc. — **Menlo Park, CA, USA**
Research Scientist, AI system Co-design — *May 2026 – Now*
Conduct research on hardware performance modeling and optimization on large-scale GenAI and recommendation models, including GPUs and Meta in-house AI hardware.

University of Michigan — **Ann Arbor, MI, USA**
Graduate Research Assistant, Advisor: Reetuparna Das — *Sep. 2020 – Apr. 2026*
Co-design architecture and system for large-scale emerging applications, including Generative AI and genome sequencing. Publish papers at top conferences including ISCA, ASPLOS and HPCA, awarded as Communication of ACM Research Highlights (24/10,000+) and HPCA Distinguished Artifact Honorable Mention (3/29).

Tenstorrent Inc. — **Santa Clara, CA, USA**
Performance Architect Intern, Manager: Wei-han Lien — *May 2023 – Aug. 2023*
Build a profile-guided agile performance model for Tensix AI accelerator, including computation and communication between RISC-V cores, with $<10\%$ error rate. This model reduces the performance modeling time from a few hours to $\sim$5 minutes, enabling fast kernel development and optimization.

Intel Labs — **Hillsboro, OR, USA**
Graduate Research Intern, Manager: Nilesh Jain — *June 2022 – Aug. 2022*
Develop an offline profile and online prediction-based resource scheduler for Quality of Service (QoS)-aware job co-location for ML inference, achieving up to $1.4\times$ request throughput improvement over single model deployment and reducing scheduling times from $\sim$30 minutes to $\sim$10 seconds compared to state-of-the-art schedulers.

Yale University — **New Heaven, CT, USA**
Undergraduate Research Intern, Advisor: James Duncan, Xiaoxiao Li — *Nov. 2019 – Mar. 2020*
Interpretation on ASD with fMRI and Deep Learning Models. Publications on MICCAI and MedIA.

École Polytechnique Fédérale De Lausanne (EPFL) — **Lausanne, Switzerland**
Undergraduate Research Intern, Advisor: Babak Falsafi — *July 2019 – Sep. 2019*
Patched QFlex Simulator (from Makefile to CMake compilation) and accelerated activation functions for DNN.

Award & Honors

Outstanding Young Speaker Award (APPT 2025) — *July 2025*
Hardware-Software Co-Design for Scalable LLM Inference

Distinguished Artifact Honorable Mention (HPCA 2025) — *March 2025*
Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing. (3/29 Artifacts)

Communication of ACM (CACM) Research Highlights — *July 2024*
GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis.
CACM Research Highlights section selects 24/10,000+ papers from ACM conferences per year, reprinting the most significant and influential results across Computer Science.

University of Michigan Rackham Graduate Student Research Grant ($3000, role: PI) *Apr. 2024*
Pangenomics Benchmark Suite and Characterization.

University of Michigan Rackham Travel Grant, ISCA/HPCA Travel Grant *2023, 2025*

Fellowship of Summer@EPFL (2% applicants awarded) *July 2019*

Tang Lixin Fellowship (2‰ students awarded) *Nov. 2017, 2018, 2019*

Outstanding Student Leaders in Zhejiang University (3% students awarded) *Oct. 2017, 2019*

First-Class Scholarship for Outstanding Students (2% students awarded) *Oct. 2017*

Under-Review Publications

Yufeng Gu, Reetuparna Das. "Adaptive Scheduling for Large Language Model Inference on Heterogeneous Server Farm *(Under Submission to ASPLOS'27)*

Yufeng Gu, Samel Gobriel, Nilesh Jain, Reetuparna Das. "BRIDGE: A Systolic Processing-In-Memory Architecture for Narrow GEMMs in LLM Inference" *(Under Submission to IEEE Transactions on Computers)*

Yufeng Gu, Vui Seng Chua, Nilesh Jain, Ravishankar Iyer, Reetuparna Das. "Fast Resource Management for Quality of Service-Aware Co-Location of Machine Learning Inference." *(Under Submission to SC'2026)*

Sumanth Umesh, Ning Liang, **Yufeng Gu**, Reetuparna Das. "Addressing In-Memory Database Bottlenecks with CXL Memory Expansion." *(Under Submission to VLDB)*

Peer-Reviewed Publications

* equal contribution

Jiajun Luo, Yu Jin, **Yufeng Gu**, Jinyi Deng, Yang Hu and Shuwen Deng. "Stone-Skimming: Attacking Big-Data Analytics Applications with Page Table Walk." To appear in 63rd ACM/IEEE Design Automation Conference (DAC'26).

Noah Kaplan, Jan-Niklas Schmelzle, **Yufeng Gu**, Christopher Batten, Reetuparna Das. "PangenomicsBench: A Benchmark Suite and Characterization of Pangenomics." In IEEE International Symposium on Workload Characterization (IISWC'25).

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." Communications of the ACM, 2025.

Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." In Proceedings of the 52th Annual International Symposium on Computer Architecture (ISCA'25).

Yufeng Gu*, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25). **In Progress of Technology Transfer to SK Hynix's Next Generation Datacenter**

Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." In 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**

Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'21).

Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. *IF = 11.1*

Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI'20).

Talks

Architecture and System Co-Design for Scalable LLM Inference
Chinese University of Hong Kong (CUHK) — *Dec. 2025*
PhD Forum at APPT 2025 — *July 2025*
Los Alamos National Laboratory — *May 2025*

CXL-enabled PIM system for LLM inference
SAFARI Live Seminar (Host: Prof. Onur Mutlu) — *June 2025*
MCCSys workshop co-located with ASPLOS 2025 — *Mar. 2025*

Genomics Benchmark Suite and Accelerator Design
Computer Architecture Seminar at UCF (Host: Prof. Di Wu) — *Mar. 2024*
Cornel University (Host: Prof. Christopher Batten) — *Feb. 2024*
Peisu Xia Forum at ICT, CAS — *Dec. 2023*

Teaching

Guest Lecture at UMich EECS 570 Parallel Computer Architecture (Fall 2025 and Winter 2026)

Programming at Discovery Engineering program at University of Michigan — *July. 2022*

Mathematics at Tuanlin Primary School, Guizhou, China — *Aug. 2017*

Services

Reviewer for IEEE Computer Architecture Letter (CAL), IEEE Transaction on Computer (TC), IEEE Transaction on Big Data (TBD), Journal of Systems Architecture (JSA).

Artifact evaluation reviewer for IISWC 2021, ISCA 2023, ISCA 2024, ISCA 2025.

Mentorship

Linus Wang B.S. UM → Ph.D. at UMich — *Jan. 2025 - Apr. 2026*

Wenjie Geng M.S. UM — *Sep. 2024 - Aug. 2026*

Zesen Zhao M.S. UM → Ph.D. at UMich — *Jan. 2025 - Aug. 2025*

Mayne Mei B.Eng. UM → Engineer at Etched — *Jan. 2024 - Apr. 2024*

Yuzhe Ruan B.S. UM → M.S. at Yale — *Sep. 2023 - Apr. 2024*

Donglin Yu B.Eng. → M.Eng. at UIUC — *Jan. 2023 - Apr. 2024*

Dawit Melka B.Eng. Addis Ababa University → Engineer at iCog Labs *May 2022 - Aug. 2022*

James Gu B.Eng. UM → Engineer at Amazon *May 2022 - Aug. 2022*